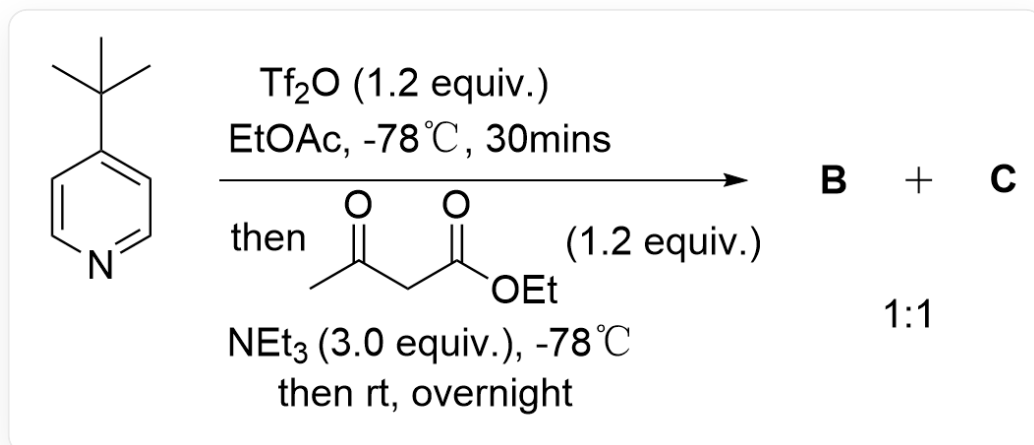


## 题目

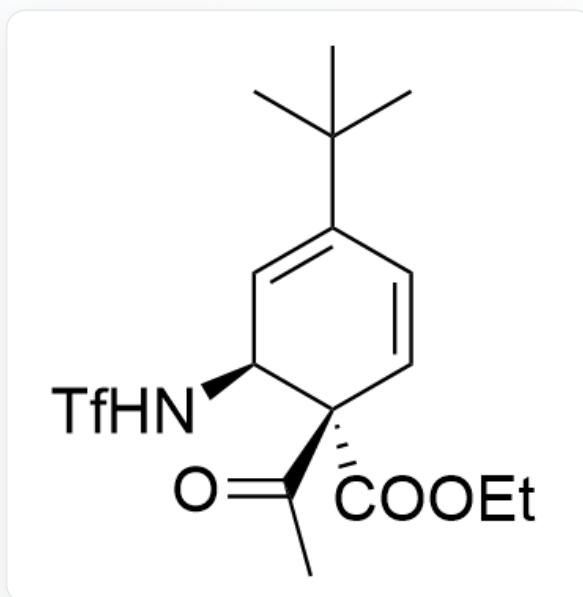


CC(C)(C)C1=CC=NC=C1 >  $\text{Tf}_2\text{O}$  (1.2 equiv.),  $\text{EtOAc}$ , 在  $-78^\circ\text{C}$  反应 30 mins; then CC(CC(OCC)=O)=O (1.2 equiv.),  $\text{NEt}_3$  (3.0 equiv.), 在  $-78^\circ\text{C}$ ; then rt, overnight > [**B**].[**C**], 产物 **B** 和 **C** 的比例约为 1:1

已知该反应得到产物 **B** 和 **C** 的比例大致为 1:1 且二者分子式不同，产物 **C** 的分子式为  $\text{C}_{16}\text{H}_{22}\text{F}_3\text{NO}_5\text{S}$ ，不考虑对映异构的条件下，试给出产物 **C** 的结构式

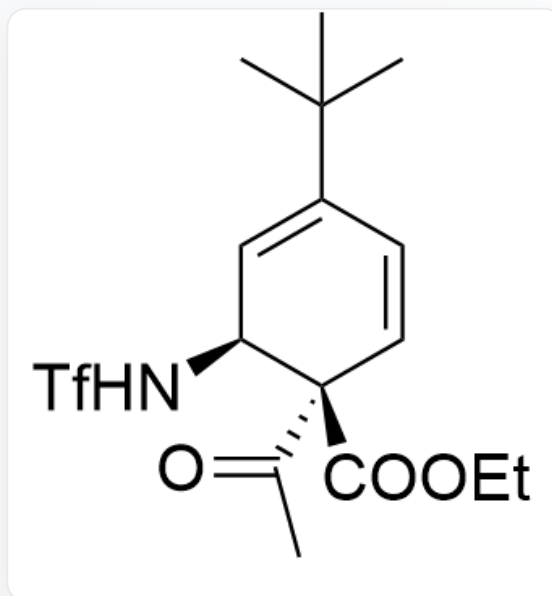
**A.** 其他选项均不正确

B.



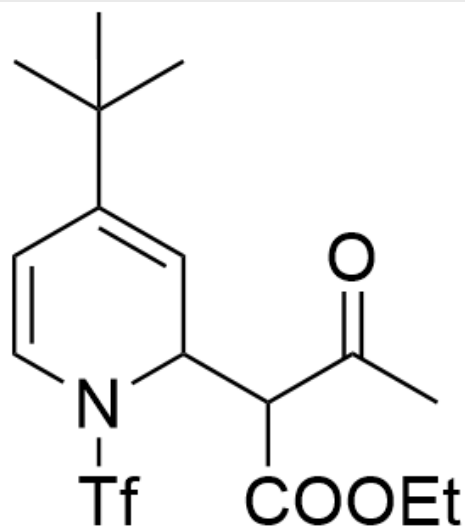
CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@@](C(C)=O)(C(OCC)=O)C=C1

C.



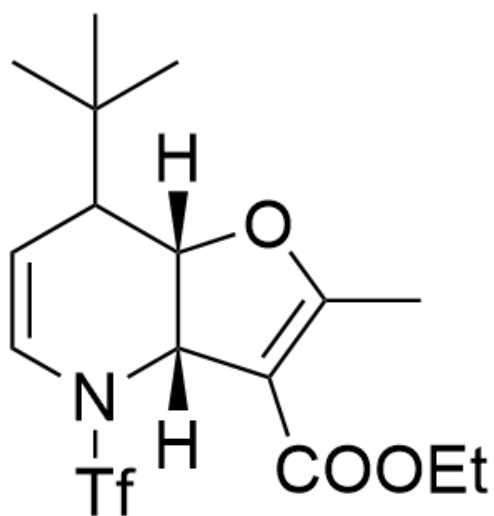
CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@](C(C)=O)(C(OCC)=O)C=C1

D.



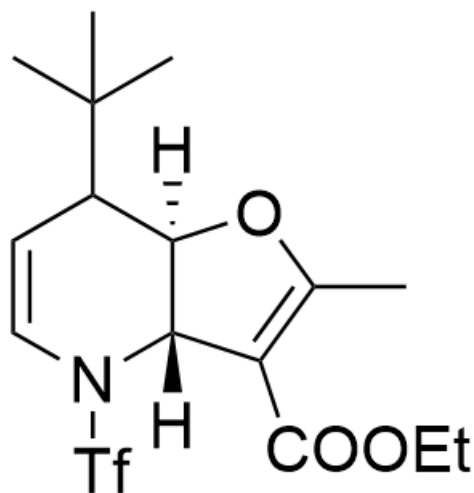
CC(C)(C)C1=CC(C(C(=O)OCC)=O)C(C)=ON(S(=O)(=O)C(F)(F)F)=O)C=C1

E.



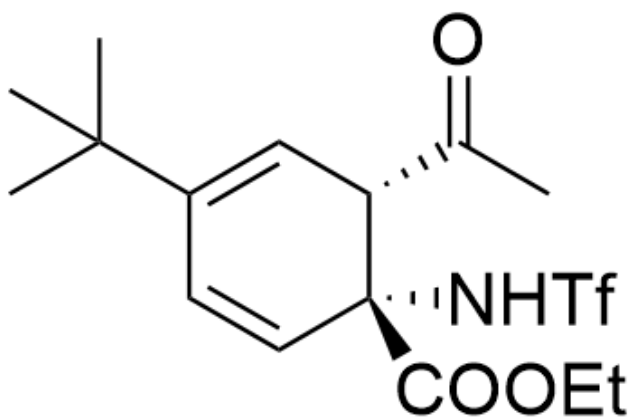
CC(C)(C)C1[C@@]2([H])[C@@](C(C(=O)OCC)=O)([H])N(S(=O)(=O)C(F)(F)F)=O)C=C1

F.



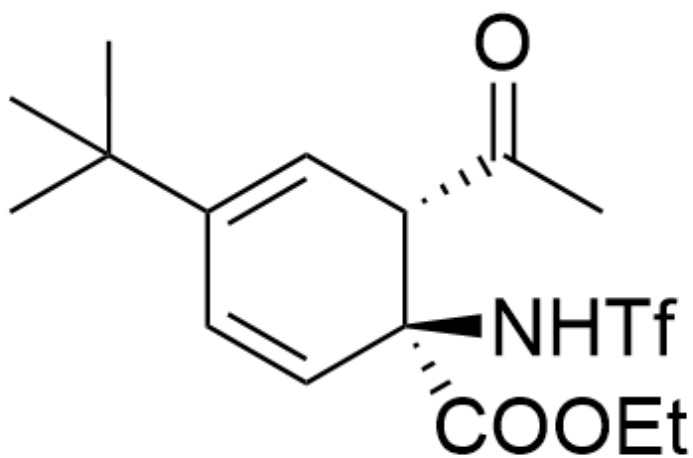
CC(C)(C)C1[C@]2([H])[C@@](C(C(OCC)=O)=C(C)O2)([H])N(S(=O)(C(F)(F)F)=O)C=C1

G.



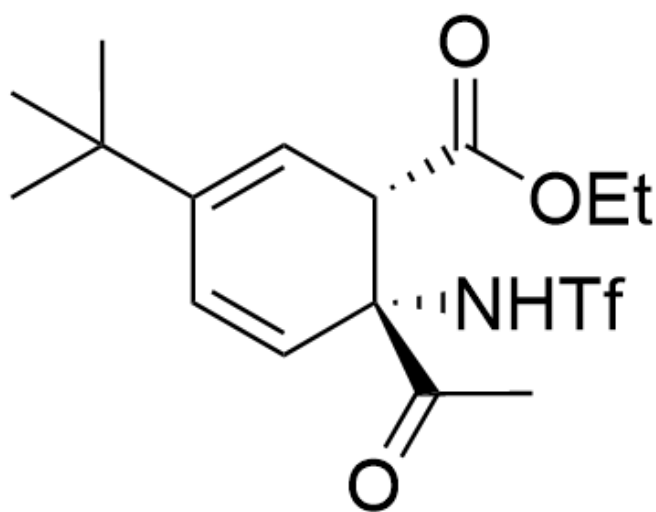
CC(C)(C)C1=C[C@H](C(C)=O)[C@](C(OCC)=O)(NS(=O)(C(F)(F)F)=O)C=C1

H.



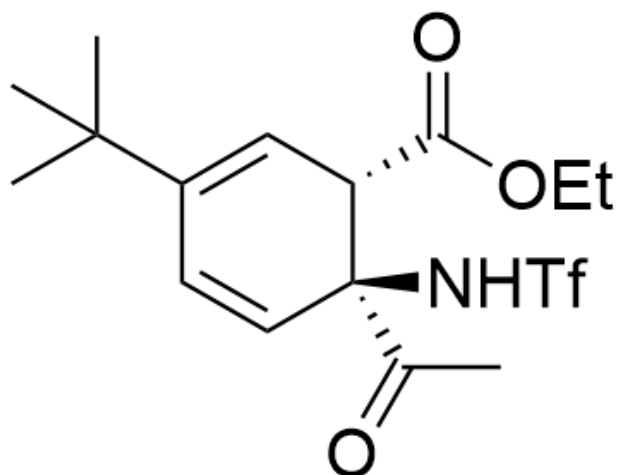
CC(C)(C)C1=C[C@H](C(C)=O)[C@@](C(OCC)=O)(NS(=O)(C(F)(F)F)=O)C=C1

I.



CC(C)(C)C1=C[C@H](C(OCC)=O)[C@](C(C)=O)(NS(=O)(C(F)(F)F)=O)C=C1

J.



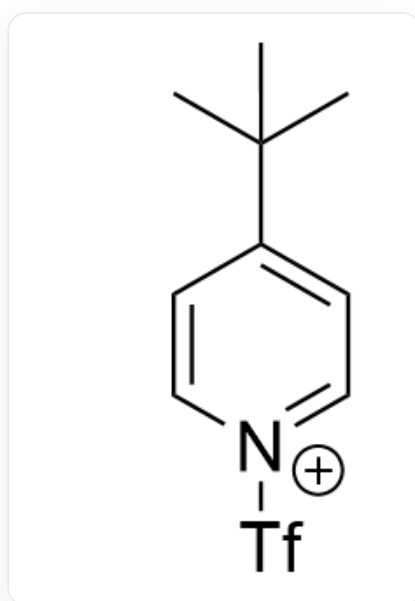
CC(C)(C)C1=C[C@H](C(OCC)=O)[C@@](C(C)=O)(NS(=O)(C(F)(F)F)=O)C=C1

## 答案

正确答案: C

## 详细解析

首先观察第一步反应，显然是底物与  $\text{ Tf}_2\text{O}$  发生反应得到中间体 1



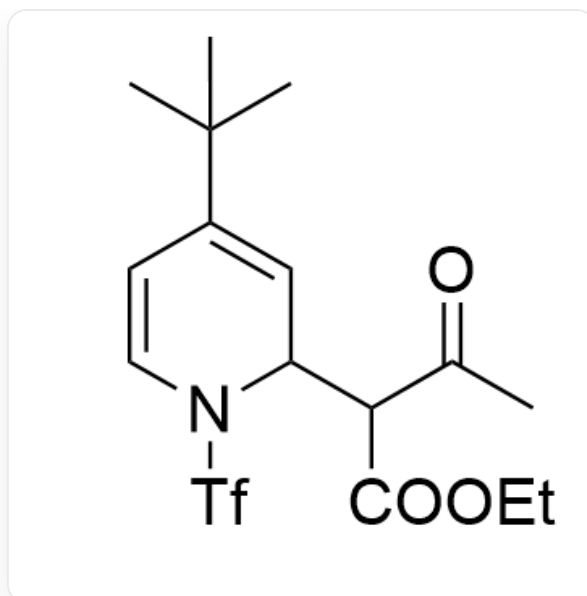
中间体1: CC(C)(C)C1=CC=[N+](S(=O)(C(F)(F)F)=O)C=C1

### CHECKPOINT

1 PTS

中间体 1: CC(C)(C)C1=CC=[N+](S(=O)(C(F)(F)F)=O)C=C1

接着第二步发生加成反应得到中间体 2



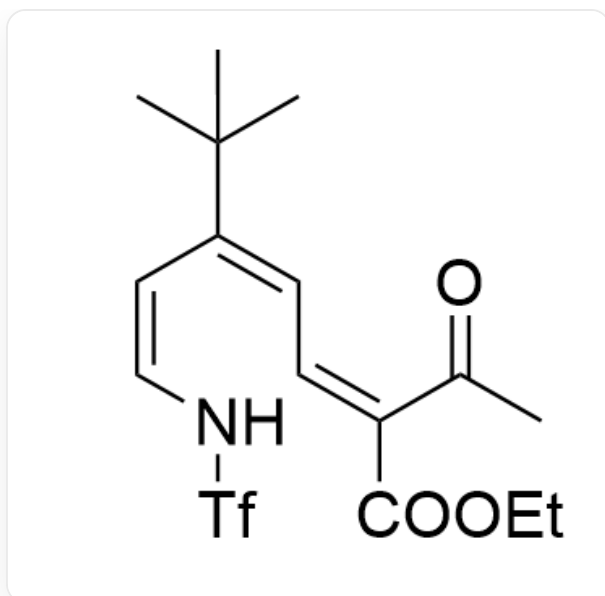
中间体**2**: CC(C)(C)C1=CC(C(C(OCC)=O)C(C)=O)N(S(=O)(C(F)(F)F)=O)C=C1

### CHECKPOINT

1 PTS

中间体**2**: CC(C)(C)C1=CC(C(C(OCC)=O)C(C)=O)N(S(=O)(C(F)(F)F)=O)C=C1

在过量碱催化作用下，进一步发生消除反应得到中间体**3**



中间体**3**: CC(C)(C)C(/C=C\NS(=O)(C(F)(F)F)=O)=C/C=C(C(OCC)=O)\C(C)=O

#### CHECKPOINT

1 PTS

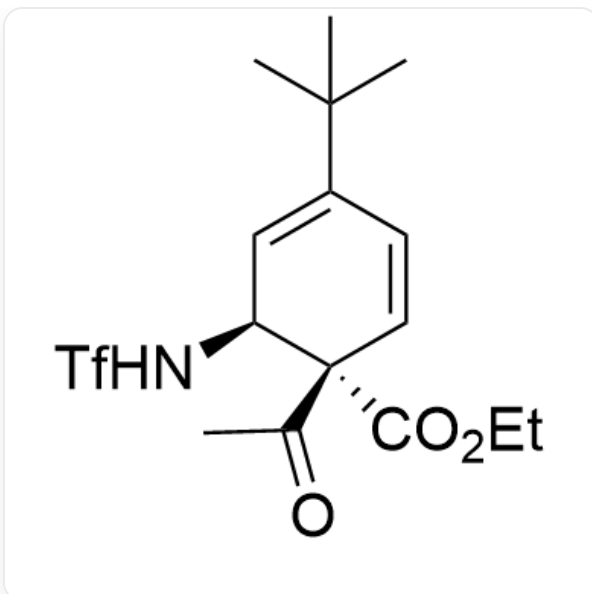
中间体 **3**: CC(C)(C)C(/C=C\NS(=O)(C(F)(F)F)=O)=C/C=C(C(OCC)=O)\C(C)=O (其中双键顺反不要紧)

接着发生六电子电环化生成一对非对映异构体，此时产物比例恰好大致约为1:1，但是两者分子式依然相同

#### CHECKPOINT

1 PTS

接着发生六电子电环化生成一对非对映异构体，此时得到产物比例恰好大致约为1:1，但是两者分子式依然相同

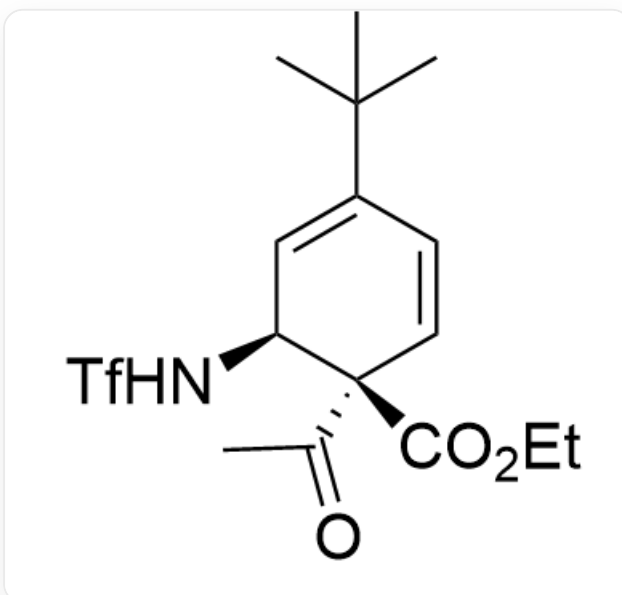


中间体4: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@@](C(C)=O)(C(OCC)=O)C=C1

CHECKPOINT

1 PTS

中间体4: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@@](C(C)=O)(C(OCC)=O)C=C1



中间体5: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@](C(C)=O)(C(OCC)=O)C=C1

CHECKPOINT

1 PTS

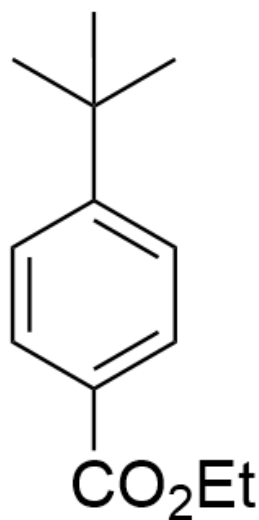
中间体 **5**: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@](C(C)=O)(C(OCC)=O)C=C1

此时由于酮相对于酯的亲电性更强，因此只有中间体 **4** 能进一步发生亲核-消除反应得到产物 **B**

CHECKPOINT

1 PTS

此时由于酮相对于酯的亲电性更强，因此只有中间体 **4** 能进一步发生亲核-消除反应得到产物 **B**



产物 **B**: CC(C)(C)C1=CC=C(C(OCC)=O)C=C1

## CHECKPOINT

1 PTS

产物 **B**: CC(C)(C)C1=CC=C(C(OCC)=O)C=C1

产物 **C** 即为中间体 **5**